

SMoDO - Practical 2 - Stata

Datasets required: `lbw (webuse)` and `insecticide.dta`

Introduction

In the first part of this session we will use a publicly available dataset on birthweights of babies, taken from the book *Applied Logistic Regression* by David W Hosmer and Stanley Lemeshow.

“Data were collected as part of a larger study at Baystate Medical Center in Springfield, Massachusetts. This dataset contains information on 189 births to women seen in the obstetrics clinic. Fifty nine of these births were defined as low birth weight [defined as less than 2500 grams].”

Hosmer & Lemeshow, Applied Logistic Regression (second edition, 2000), Pub. by Wiley-Interscience

In the second part of the session we will use data from a toxicological experiment. Eight groups of insects were exposed for five hours to gaseous carbon disulphide (CS_2) at different concentrations, with the purpose of investigating how the risk of death depends on the dose received.

Aims

The aims of this session are:

1. understand how to interpret results from logistic regression models.
2. understand how to compare the fits of logistic regression models (in Stata).

Part A: Investigation into low birthweights in Massachusetts, USA

We will first investigate factors which may be related to the chance of having a low birthweight baby, using the `lbw` dataset. This dataset is available directly from the Stata website, using the “webuse” command:

1. Start a new Do file, and load the data.

```
webuse lbw, clear
```

2. Explore the dataset, checking for missing values. Be sure to identify the outcome variable, and check how it is coded. Also note that the mother’s weight is given in lbs, whilst the baby’s weight is given in grams.

3. Use the `glm` command to fit a logistic model for the low birthweight variable with the mother's weight at last menstrual period (in lbs) as the single covariate.

Discuss: Interpret both of the estimated coefficients and their associated p-values and 95% confidence intervals.

4. Refit the model using the `logistic` and `logit` commands, and compare the output. What are the main differences between the commands and the output that they produce?
5. Using pen and paper and a calculator, calculate the probability that a women who weighs 120lbs (about 54.4kg) has a baby with birth weight less than 2500g.
6. Use Stata to construct a confidence interval around your estimate from question 5. You will need to use either the `lincom` command, or re-parameterise the model to centre weight at 120lbs.

Using `lincom`:

```
lincom 120*lwt + _cons, eform
```

Re-parameterise the model:

```
gen lwt120 = lwt - 120
```

```
logit low lwt120, or
```

7. Fit an appropriate model (using logistic or logit), and perform an appropriate test, to examine whether race has an independent association with the probability of having a low birthweight baby, after adjusting for the mother's weight.

Discuss: What do you conclude?

8. When the logistic model is fitted using logistic or logit, a LR `chi2( )` and `Prob > chi2` output is given in the top right hand corner. To understand what this is, perform a profile log-likelihood ratio test comparing the model you fitted in the previous part to the null model. The null model is the model which contains no covariates. What can you deduce about the meaning of the LR `chi2( )` and `Prob > chi2` values in the model you fitted in the previous part? How can you interpret the result in this case?

Part B: Investigation into effect of CS₂ at different doses on insect survival

The data are in `insecticide.dta`. There is one record per insect with `id` indexing the insect, `dosegroup` indicating which of eight dose groups the insect belonged to, `dose` giving the actual dose of insecticide received (mg/l) in the group and `y` indicating whether or not the insect died (0/1 equating to survived/died).

The aim of the analysis is to investigate the nature of the association between dose level and the proportion of insects killed?

9. Load and explore the data. Are there any missing values? How many insects do you have data for?

10. Fit a logistic regression model relating whether or not the insect died to **dose** treating this as a continuous variable. Is there evidence of an increasing risk of death with increasing dose?
11. Interpret the dose parameter in terms of an odds-ratio, and obtain its 95% CI.
12. According to the fitted model, what is the estimated probability of death at a dose of 55 mg/l CS₂?
13. Work out the dose that, on the basis of this fitted model, would lead to 50% of the insects dying (sometimes this is termed the LD50).
14. Calculate a new variable equal to the square of the dose. Use this to investigate whether the inclusion of a quadratic dose term improves the fit of the linear dose model. Also plot the predicted probabilities of death from this model against dose, contrasting these predictions with those from the model without the quadratic term.

Note that Stata's command for obtaining predicted probabilities is (after fitting a model) `predict <newvar>`.

15. Now consider dose as a categorical variable. Fit a logistic regression model relating whether or not the insect died to **dosegroup** treated as a categorical variable.

Discuss: Explain why the earlier model where dose was treated as a continuous predictor is nested in this one

16. Carry out an appropriate likelihood ratio test to compare the fits of the models where dose is considered as continuous and categorical. What do you conclude?